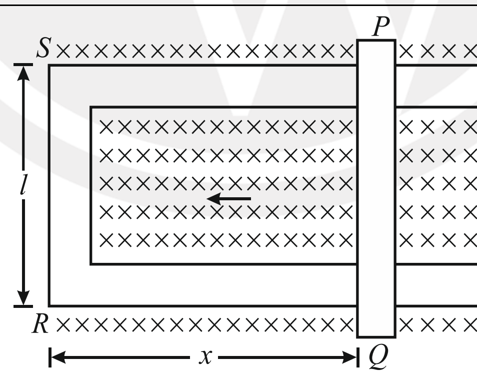
CHAPTER 6

Electromagnetic Induction

Sr. No.	Concept	Formulas	Other information
1.	**Magnetic flux**	The magnetic flux (ϕ) $$\phi = BA\cos\theta = \vec{B}\cdot\vec{A}$$	θ = angle between $\vec{A}$ and $\vec{B}$.
2.	**Faraday's Laws of electromagnetic induction**	Mathematically, $E \propto \frac{d\phi}{dt}$ $$\Rightarrow E = -\frac{d\phi}{dt}$$ If a coil has N number of turns $$E = -N\frac{d\phi}{dt}$$	$d\phi$ = change in magnetic flux in time interval dt. E = emf induced
3.	**Motional emf (Induced emf)**	 Then the magnetic flux linked with the rectangular loop $PQRS$ is $\phi = BA = Blx$ $$E = -\frac{d\phi}{dt} = -\frac{d}{dt}(Blx) = -Bl\frac{dx}{dt} \text{ or } \boxed{E = Blv}$$	B = magnetic field of the loop l = length of the coil. A = area of the loop. v = velocity of movable side PQ.

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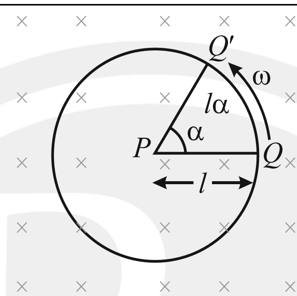
4.	**Power delivered by the external force and dissipated power in a conductor moved in magnetic field**	Power delivered by the external force: $$P = Fv = \frac{B^2l^2v^2}{R}$$ The power dissipated in the rod as Joule heating loss is $$P_J = I^2R = \left(\frac{Blv}{R}\right)^2 R = \frac{B^2l^2v^2}{R}.$$	P = power required to move a rod (PQ) of length l. B = magnetic field. v = velocity of the loop. R = resistance of the movable arm. (Refer to the diagram in section 3)
5.	**Motional emf of a conducting rod rotating in a magnetic field.**	 Flux linked, $\phi = BA\cos 0^\circ = BA$ $$= B \times \frac{1}{2}l^2\alpha = \frac{1}{2}B\omega l^2\alpha$$ Induced emf in the rod: $$\varepsilon = \frac{1}{2}Bl^2\omega$$	l = length of rod PQ. ω = angular velocity. B = magnetic field. α = the rod turns by an angle in time t. A = area swept.
6.	**Induced current and induced charge (circular loop)**	If R is the resistance of the circuit, then induced current is given by $$i = \frac{E}{R} = \frac{1}{R}\left(\frac{-d\phi}{dt}\right)$$ If induced current is produced in a coil rotated in a uniform magnetic field, then $$E = -\frac{d\phi_B}{dt} = \frac{-d}{dt}(NBA\cos\omega t)$$ $$= NBA\omega\sin\omega t$$ $$I = \frac{NBA\,\omega\sin\omega t}{R} = I_0\sin\omega t$$ where, $I_0 = \frac{NBA\,\omega}{R}$ = peak value of induced current.	N = number of turns in the coil. B = magnetic field. ω = angular velocity of rotation. A = area of cross-section of the coil.

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7.	**Self-induction of coil (Solenoid and Toroid)**	The magnetic flux linked with a coil. $$\phi = LI$$ The induced emf in the coil, $$E = -L\frac{dI}{dt}$$ Self-inductance of a long solenoid is given by $$L = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 Al$$ If core of the solenoid is of any other magnetic material, then $$L = \frac{\mu_0 \mu_r N^2 A}{l}$$ Self-inductance of a toroid, $L = \frac{\mu_0 N^2 A}{2\pi r}$ Energy stored in an inductor, $E = \frac{1}{2}LI^2$	L = coefficient of self-induction. N = total number of turns in the solenoid. l = length of the coil. n = number of turns per unit length in the coil. A = area of cross-section of the coil. r = radius of the toroid. μ_r = relative permeability of the core material.
8.	**Mutual induction (two coils are coupled)**	If two coils are coupled with each other then magnetic flux linked with a coil (secondary coil) $$\phi = MI$$ The induced emf in the secondary coil, $$E = -M\frac{dI}{dt}$$ Mutual inductance of two long coaxial solenoids is given by $$M = \frac{\mu_0 N_1 N_2 A}{l} = \mu_0 n_1 n_2 Al$$ A is area of cross-section of solenoids and l is length of the solenoids.	M = coefficient of mutual induction. I = current flowing through primary coil. ϕ = magnetic flux associated with secondary coil. N_1 and N_2 = total number of turns in primary and secondary coil respectively. n_1 and n_2 = number of turns per unit length in primary and secondary coil.

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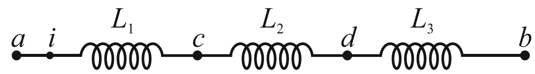
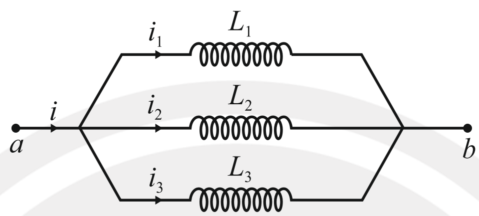
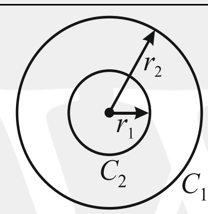
9.	**Grouping of Inductances (Equivalent Self-inductance)**	(i) In case of three inductors connected in series:  Then, their equivalent inductance is given by $\boxed{L = L_1 + L_2 + L_3}$ (ii) In case of three inductors connected in parallel:  Then, their equivalent inductance L is given by $\boxed{\frac{1}{L} = \frac{1}{L_2} + \frac{1}{L_2} + \frac{1}{L_3}}$	L_1, L_2 and L_3, = three inductors connected in series and parallel respectively. K = coefficient of coupling.
10.	**Mutual inductance of two concentric coils of different radius**	 Magnetic flux linked with coil C_1: $\phi_B = B_2 A_1 = \left(\frac{\mu_0 I}{2r_2}\right)(\pi r_1^2) = \frac{\mu_0 \pi r_1^2}{2r_2} I$ Mutual inductance between two coils: $M = \left(\frac{\mu_0}{4\pi}\right)\frac{2\pi^2 r_1^2}{r_2}$	r_1 and r_2 = radius of the two rings C_2 and C_1 respectively. I = current flowing through the coil C_1 of radius r_2.
11.	**Coefficient of coupling**	Coefficient of coupling $K = \frac{M}{\sqrt{L_1 L_2}}$ The value of K lies between 0 and 1.	L_1 and L_2 = self-inductance of two coils. M = mutual inductance.

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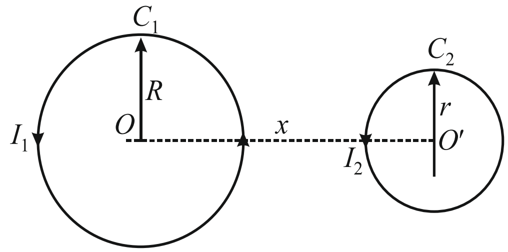
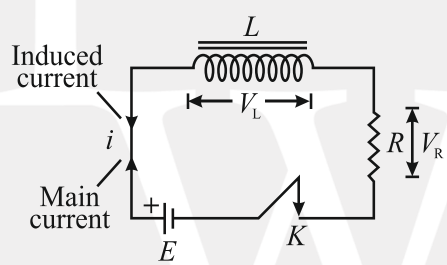
12.	**Mutual inductance of two co axial coils of different radius (*M*)**	 Magnetic flux linked with coil C_2: $\phi = B_1 A_2 = \frac{\mu_0}{4\pi} \frac{2\pi R^2 I_1}{(R^2+x^2)^{3/2}} \times \pi r^2$ $= \frac{\mu_0}{4\pi} \frac{2\pi^2 R^2 r^2}{(R^2+x^2)^{3/2}} I$ Mutual Inductance between two coils: $M = \left(\frac{\mu_0}{4\pi}\right) \frac{2\pi^2 R^2 r^2}{(R^2+x^2)^{3/2}}$	R and r = radius of two co-axial coils C_1 and C_2. I_1 = current flowing through the coil C_1. M = mutual inductance between them.
13.	**Growth and decay of current in *L*–*R* circuit**	Growth of current:  $i = i_0\left[1 - e^{-\frac{R}{L}t}\right]$ where, $i_0 = i_{max} = \frac{E}{R} =$ steady state current at $t = \infty$ and time constant, $\tau_L = L/R$. So, the Eq. can be re-written as $i = i_0(1 - e^{-t/\tau_L})$ Decay of current: $i = i_0 e^{-\frac{R}{L}t} = i_0 e^{-t/\tau_L}$	R = external resistance connected with the circuit. E = emf of the cell. L = inductance of the inductor.